Susana Constenla-Villoslada

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EDUCATION

PhD – School of Information, University of California, Berkeley 2022-2026 (expected)
Advisor: Joshua Blumenstock

Multidisciplinary program (economics/computer science).
Specializing in the intersection of econometric and machine learning methods for international development research.
Inclusion of geospatial and other alternative secondary data into traditional economics research.

Main dissertation research:
- Leading the development and deployment, in direct partnership with the government of Kenya, of a machine learning based early warning system for malnutrition crises among children under five in ASAL Kenya.
- Designing and carrying the tool's deployment under a quasi-experimental impact evaluation setup.
- Development of minimum sample estimation approach for ML model retraining after distribution shift in survey-centric applications.

Cornell University –Charles H. Dyson School of Applied Economics and Management

M.S. - Applied Economics and Management (spec.: Development Economics), under Fulbright Scholarship 2017-2019

Thesis advisor: Christopher B. Barrett; Minor chair: John Hoddinott

Polytechnic University of Madrid 2011-2016

B.S. - Agricultural Engineering, with Honors

RESEARCH

PUBLICATIONS

Constenla-Villoslada S, Liu Y, McBride L, Ouma C, Mutanda N, Barrett CB. (2025). High-frequency monitoring enables machine learning-based forecasting of acute child malnutrition for early warning. *PNAS.* U S A.

Constenla-Villoslada, S., Liu, Y., Wen, J. et al. (2022). Large-scale land restoration improved drought resilience in Ethiopia's degraded watersheds. *Nature Sustainability*. https://doi.org/10.1038.

Upton, J., Constenla-Villoslada, S. and Barrett, C.B. (2022). "*Caveat utilitor*: A comparative assessment of resilience measurement approaches", *Journal of Development Economics*, Vol- 157, 102873. [Seniority shared between first two authors.]

WORKING PAPERS

Constenla-Villoslada, S., Ougo, J., Mutanda, N., Barrett, C.B., Blumenstock, J. E. Machine learning forecast-based triggers for anticipatory action on child wasting: an impact evaluation in arid Kenya.

Constenla-Villoslada, S. A minimum sample size estimation approach for retraining survey-dependent machine learning regression models after distribution shift.

FUNDING

Jameel Observatory (University of Edinburgh, J-PAL, Save the Children, ILRI) **[$75,000]** 2024

- Funding for implementation of dissertation project "Improved Early Warning of Food Insecurity"

Google.org [$250,000] 2024

- Research gift to fund work on dissertation project related to food security (implementation of early warning system for child malnutrition crises in ASAL Kenya)

NASA **[$498,974]** (under review)

- Advancing Anticipatory Action Using Improved Geospatial and /Weather Data: funding to include new Privthi Foundation Model geospatial and climate/weather predictions into the developed early warning system in ASAL Kenya.

OTHER RESEARCH EXPERIENCE

UNIVERSIDAD POLITÉCNICA DE MADRID 07/2015 – 07/2016

Senior Thesis – Agricultural Engineering

I proposed and put in practice with minimal supervision a one-year long research project to analyze possible pollution from heavy metals affecting urban orchards of the city of Madrid. The movement of urban agriculture was steadily expanding in those years in Madrid, with an increasing number of people consuming the harvested products. Assessing the potential risk of consuming those crops, produced in proven air-polluted environments, was the question that I explored. I lead the whole research process, from designing the experiments, to carrying out all the field work and laboratory analysis of the materials, to analyzing the results and producing a written piece that I presented to the tribunal (my work was evaluated by an external panel of professors and received a score of 9.2/10). I got a proposal to continue that research project, but I decided to go pursue graduate studies in the USA instead.

WORK EXPERIENCE

International Food Policy Research Institute – Research Analyst II - Washington D.C. 2022

International Food Policy Research Institute – Research Analyst I - Washington D.C. 2019-2022

Main tasks performed:

- Experimental and quasi-experimental impact evaluation design and implementation of big-scale development interventions (in Central America (Honduras); Sub-Saharan Africa (Ethiopia); South-Asia (Nepal)).
- Sample design: precision-based for population characterization, and power-based for impact evaluation purposes.
- Comprehensive dataset generation joining geospatial and socioeconomic data.
- Field experience: team training for survey data collection; field-work coordination for survey data gathering and data quality monitoring.

UNIVERSIDAD POLITÉCNICA DE MADRID 11/2016-12/2016

Agricultural engineer (internship)

I collaborated in a research project concerning the analysis of the energy efficiency of the wineries of the Spanish wine sector, to propose bioclimatic construction solutions that would help them reach minimum levels of energy consumption. I organized information from several surveys, improved the surveying method, and help to write scientific publications.

AWARDS AND FELLOWSHIPS

University of California at Berkeley

Berkeley Fellowship for PhD studies 2022-2023

Steven Weber Futures Fellowship Fund for Digital Security and Trust

Steven Weber Fellowship for Digital Security and Trust 2022

Agricultural and Applied Economics Association (AAEA)

Outstanding Master's Thesis Award (Honorable Mention) 2020

Fulbright Commission (Spain)

Full-time Fulbright scholarship to pursue graduate studies in the US 2017-2019

Polytechnic University of Madrid

Best Academic Record of the graduating class – B.S. Agricultural Engineering 2016

EXPERTISE AND TECHNICAL SKILLS

Econometrics; machine learning; statistical analysis; cross-sectional and forward-looking prediction; comprehensive, multi-source dataset generation (proficient in the rigorous intersection of geospatial and socioeconomic data), experimental and quasi-experimental impact evaluation methods.

Coding Languages: Python, Stata, R, LaTeX

Languages: English (proficient), Spanish (native), Galician (native), Portuguese (intermediate), German (basic)

VOLUNTEERING EXPERIENCE

STUDENT MULTIDISCIPLINARY RESEARCH TEAMS (SMART) – Cornell University 09/2018-05/2019

Consultant

Applied research team organized by Cornell University to conduct a service/learning project for an agricultural company ("Perisha Agro") that operates in Lilongwe, Malawi. After taking preparatory classes and gathering relevant information in the field with the company owner and different stakeholders, we developed a business plan for the company for them to apply for grants awarded by international organizations. Using the business plan we created, the company was awarded a grant from the German development agency, GIZ.